**This procedure is applicable to below models:**

| | | | | |
|---|---|---|---|---|
| V810 STD | √ | S1 | √ | S2 |
| V810 XXL | | S1 | | S2 |
| V810 S2EX | | | | |
| V810 Mini | | | | |

**REVISION HISTORY**

| Rev | Effective Date | Description | Doc Originator |
|---|---|---|---|
| 00 | 2 Oct 2016 | First Release | ONG KEAN SHENG |
| | | | |
| | | | |

## Content

## Rail Width Adjustment and Verification

1. Initialize at Panel Handling
2. Set the Display Measurement to millimeters as below Figure 1
3. Go to the Set Manual Adjustment as key in 150 and click "Set" as below Figure 1.

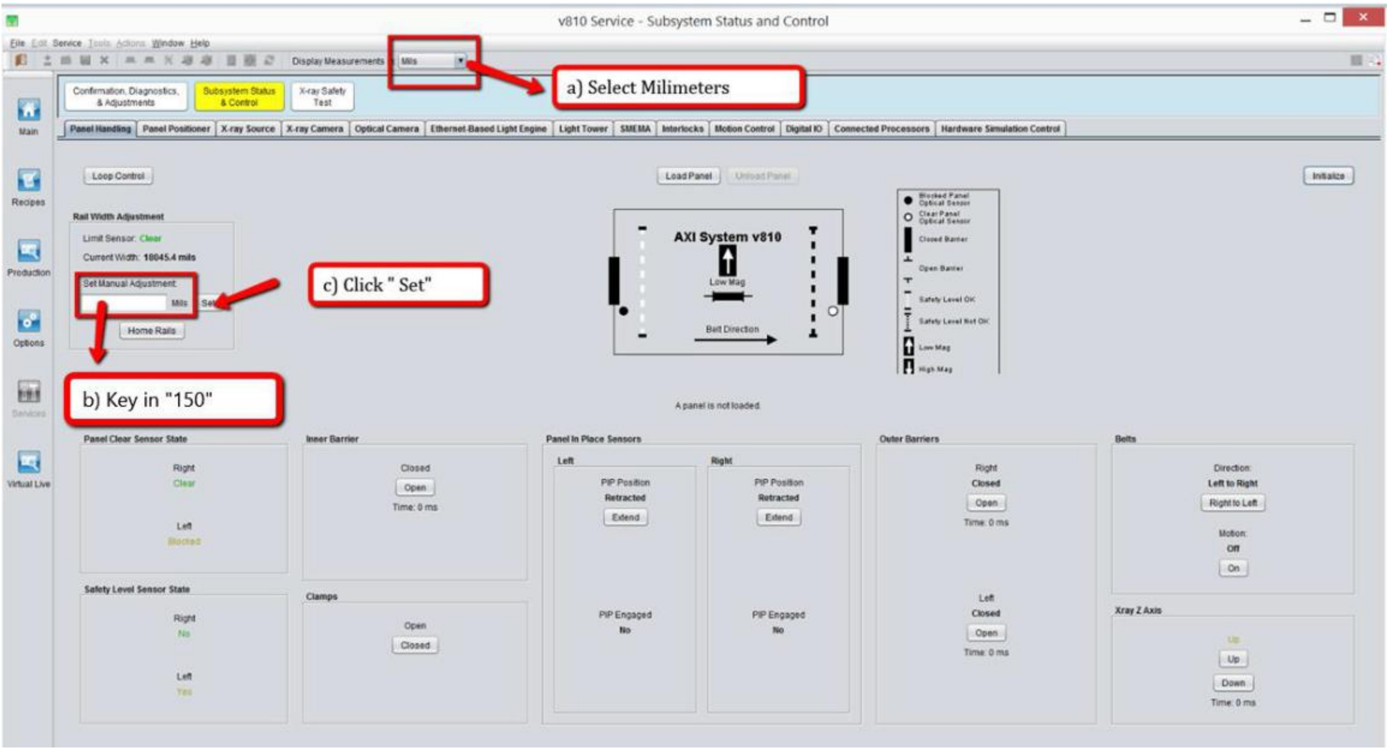



*Figure 1 Panel Handling Interface*

4. Wait until AWA is extend to desire width.
5. Close X-ray at Operator Console Panel and open the Front Door.

6. Place Caliper as below position. (May refer to below Figure 2)

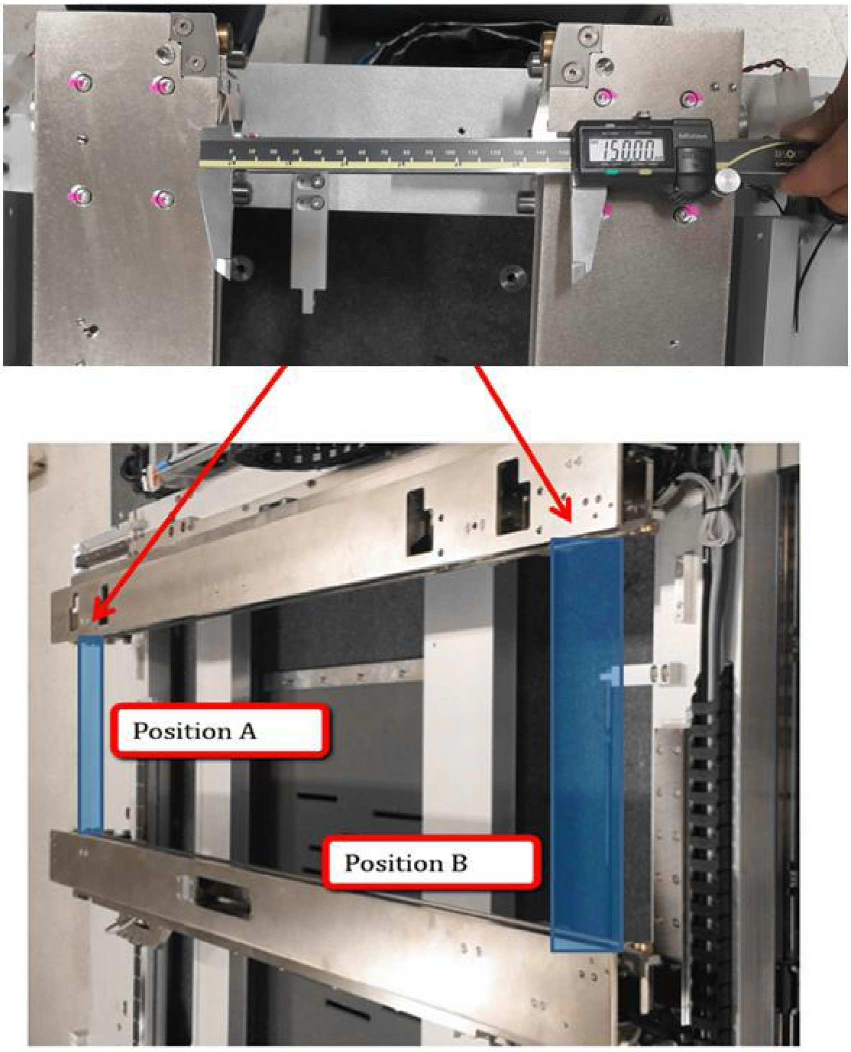
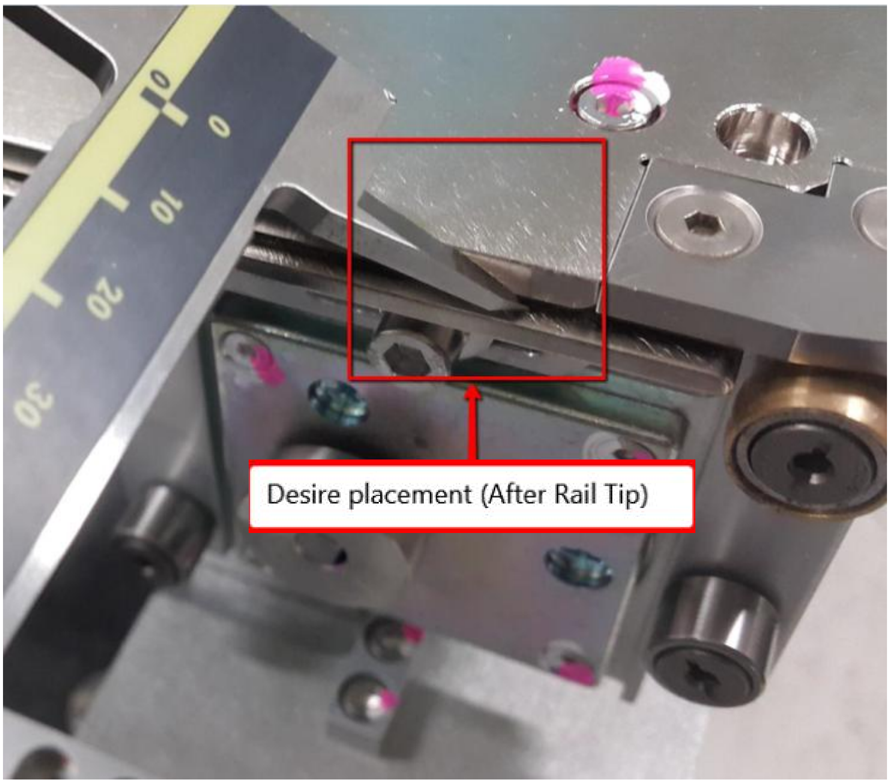
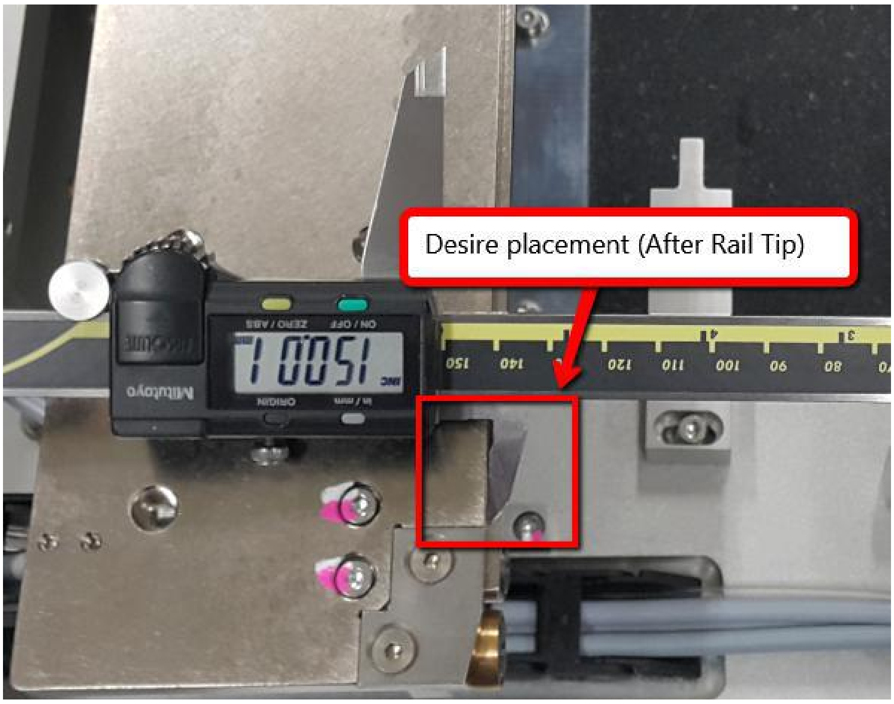



*Figure 2 Caliper Placement*

| Position | Position A & Position B |
| --- | --- |
| Fix rail Caliper placement |  |
| Adj rail Caliper placement |  |

7. Record 5 repeated measurements at the right end (Position A), and then 5 repeated measurements at the left end (Position B):
   - MRave = average value of 5 repeated measurements at right end of conveyor
   - MLave = average value of 5 repeated measurements at left end of conveyor
8. Calculate the parallelism, Mave and width error using the formulas below. Compare these values to the parallelism and width error limits.
   - Parallelism = MRave - MLave
   - Mave = (MRave + MLave)/2
   - Width Error = Set Width - Mave
   - Limits on Parallelism = +/- 0.125mm
   - Limits on Width Error = +/- 0.125mm
9. By refer to below table and step, check Parallelism Limit & Width Accuracy Limit
   - If Parallelism Limit & Width Accuracy Limit is not within spec, please refer to ***AWA Parallelism adjustment***

| Sample Measurement (mm) | | | |
|---|---|---|---|
| Measurement Right End (Position A) | | Measurement Left End (Position B) | |
| MR1 | 150.00 | ML1 | 150.10 |
| MR2 | 150.01 | ML2 | 150.08 |
| MR3 | 150.05 | ML3 | 150.08 |
| MR4 | 150.00 | ML4 | 150.06 |
| MR5 | 150.01 | ML5 | 150.09 |

Calculations:

Measurement Right End average (MRave) = (MR1+MR2+MR3+MR4+MR5)/5 = 150.01

Measurement Left End average (MLave)= (ML1+ML2+ML3+ML4+ML5)/5 = 150.08

Measurement average = (MRave + MLave)/2 = 150.04

<u>Compare to Limits for Parallelism:</u>

MRave – MLave = **0.07 is less than Spec "+/- 0.125mm ",** so no adjustment for parallelism is required

<u>Compare to Limits for Width Accuracy:</u>

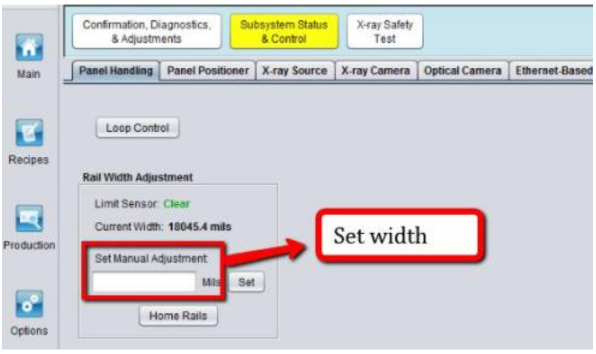



*Figure 3 Set Width*

Set Width – Mave = 0.04 is less **Spec "+/- 0.125mm ",** so no adjustment required for width accuracy

## AWA Parallelism Adjustment

1. Click "Home Rails" at panel handling. AWA will extend to maximum opening.


*Figure 4 Home Rails*

2. Open up front door, side door & rear door.

3. Use Size 8 hex wrench to loosen both side of jam nut while holding the set screw with Metrics Allen Key.

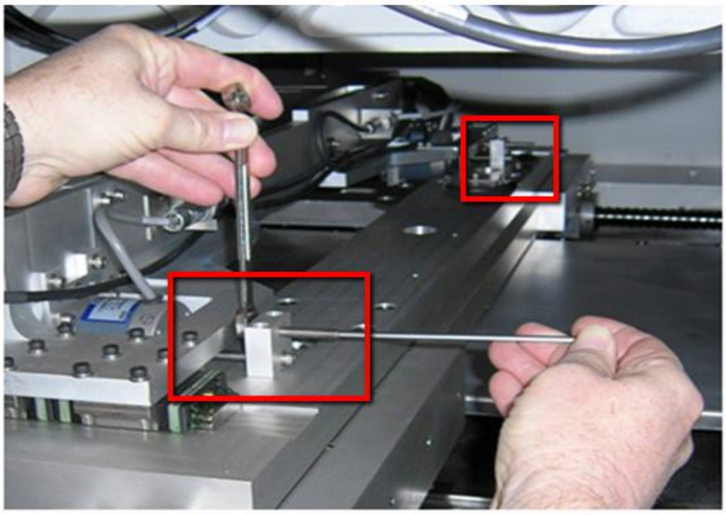


*Figure 5 Jam Nut Loosen*

4. Insert 50 mil shim (4XACM-090) plate/feeler gauge in between of inner edge of the Slide Plate X-Axis and Set Screw. Once set screw tip in contact with Shim Plate, tighten up the jam nut.

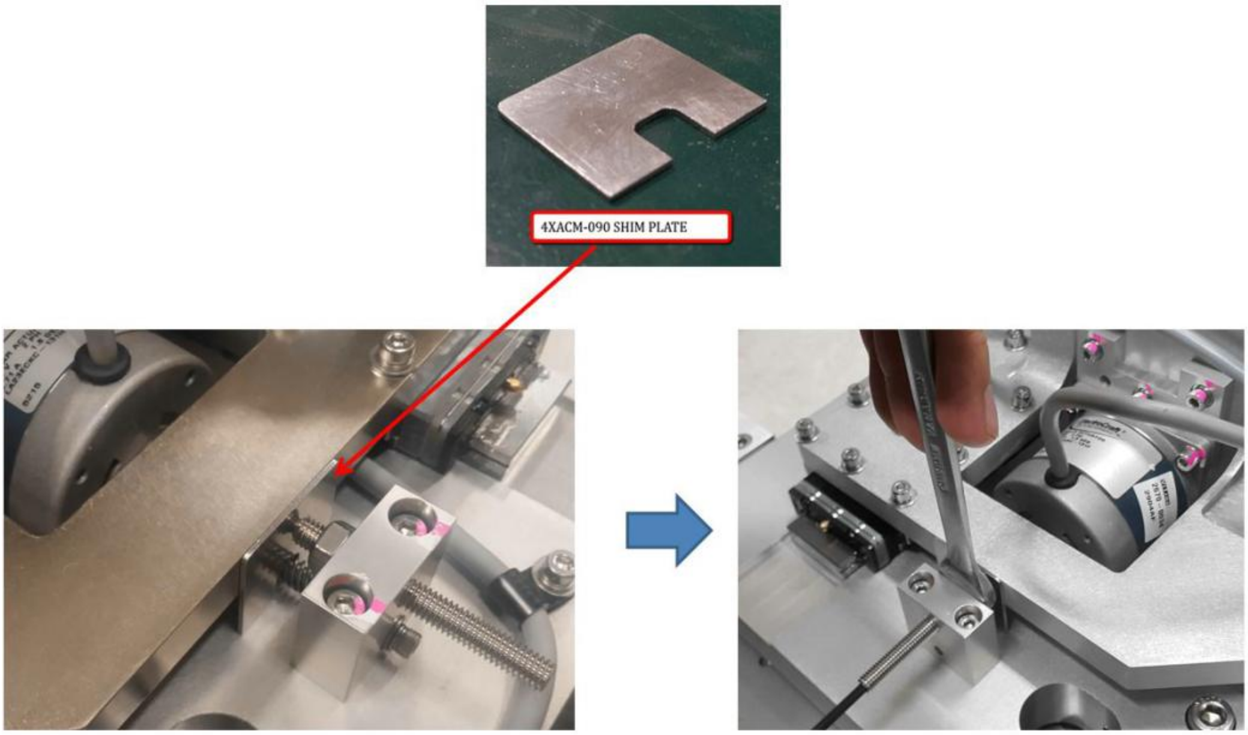



*Figure 6 50 Mil Shim Plate Insert*

5. Repeat steps 4 at other side of Adjustable rail.
6. Close SSCA.
7. Place 2 pcs of Auto Width Set Panel (4XACM-101) as below Figure. Manually push the Adjustable rail to allow both set panel fit in the desire width.

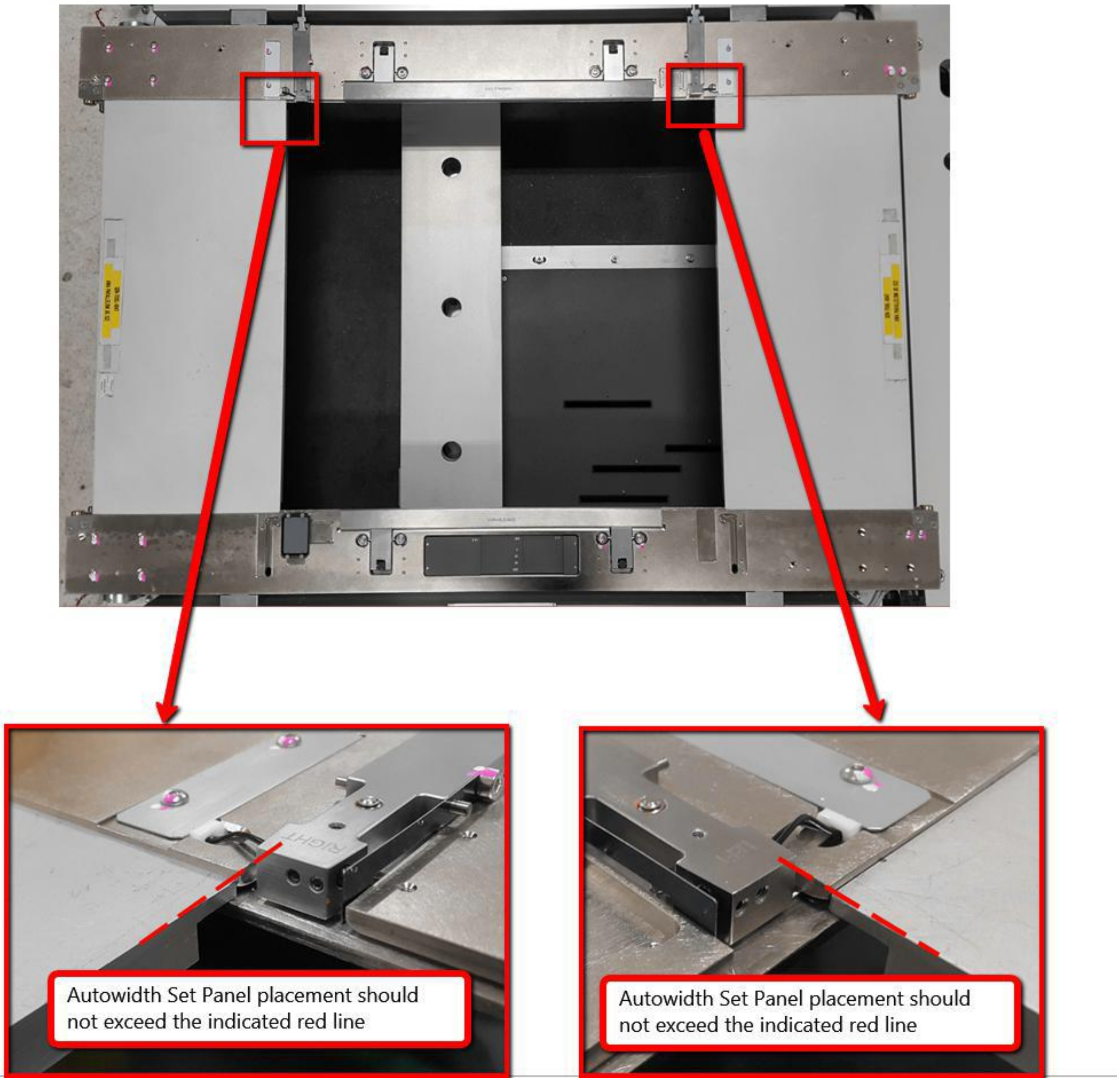



*Figure 7 Autowidth Set Panel*

8. Loosen screws at Adjustable Rail Linear Guide as below Figure 8.

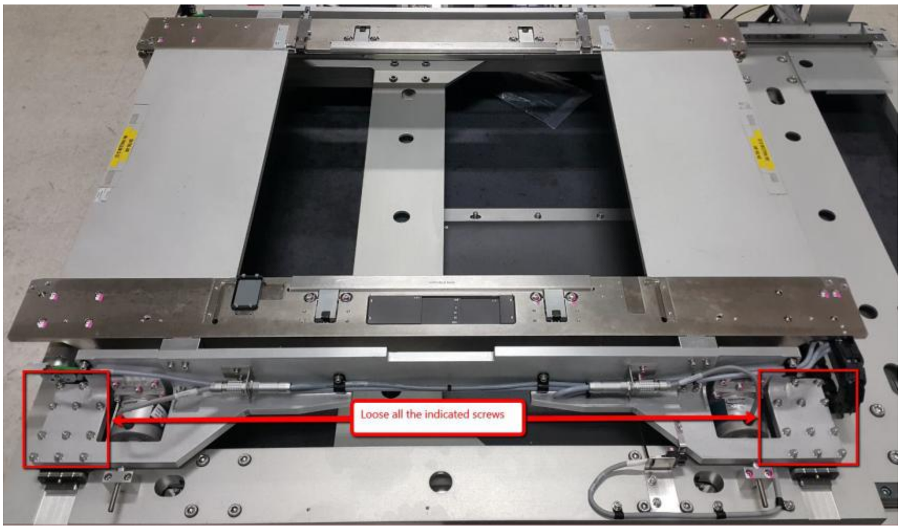



*Figure 8 Linear Guide Screws*

9. Push Left & Right Adj rail to allow Both Set Panel are fit into AWA. Tighten up both linear guide screws (Figure 9) at the same time.

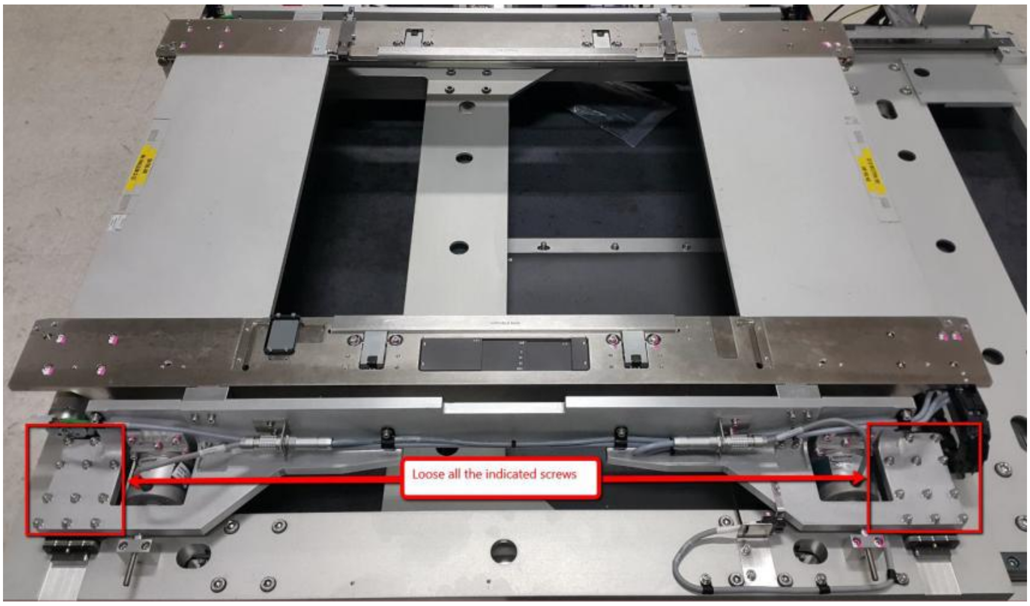



*Figure 9 AWA Linear Guide Alignment*

10. Loosen 2 x Screws at AWA Home Sensor Bracket and Set the Rail Width Home sensor to be triggered (Red Colour)

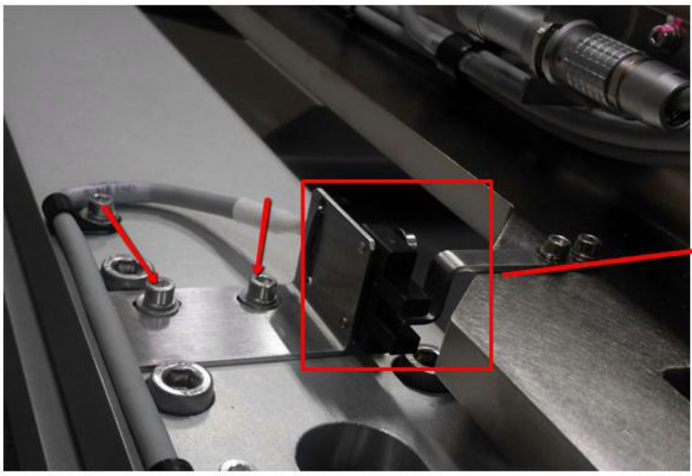
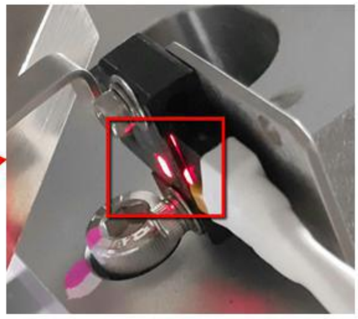



*Figure 10 AWA Home Sensor Adjustment*

11. Once Rail Width Home Sensor triggered, gently pull away the bracket towards rear side approximately 1mm.

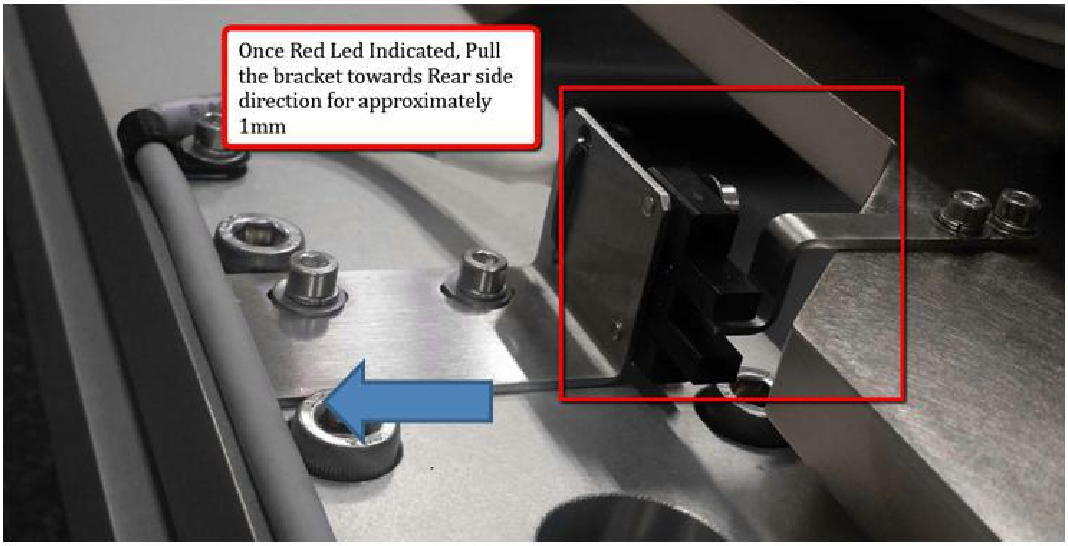



*Figure 11 AWA Home Sensor Bracket adjustment*

12. Tighten 2 x Screws at AWA Home Sensor Bracket.
13. Remove 2 x Autowidth Set Panel (4XACM-101) from rail.

14. Open back SSCA.
15. Go to Connector Processor Tab, Stop RMS & Restart RMS.
16. Initialize Digital IO.
17. Initialize at Panel Handling
18. Key in“ 150” mm at Panel handling and Click “ Set”
19. OpenHardware Config by refer to below path “C:\Program Files\AXI System\AXI System\v810\5.0\Config
20. Search the key: railWidthAxisMaximumPositionLimitInNanometers=457945000 (Default value) & record down. (Remarks : Every machine will consist varies of parameter, user will require to extract the parameter from particular machine)
21. Use the caliper to measure AWA Rail Width & record down the actual measurement.
22. By refer to below table to calculate a new value of this parameter. (Below Value are for reference only)

| Condition | Set value (Panel Handling) | Actual Measurement (Rail width) | Rail width parameter (HW config) *Step 20* | Calculation Method |
| --- | --- | --- | --- | --- |
| *If the measured value is greater than the expected value*<br><br>*(rail width is too wide),* | 150 | 155 | 457945000 | *New = 457945000 + [ (155 – 150) * 1 000 000]*<br>*New = 457945000 + (5 * 1 000 000)*<br>*New = 457945000 + 5 000 000*<br>*New = 462945000* |
| *If the measured value is less than the expected value (rail width is too narrow)* | 150 | 145 | 457945000 | *New = 457945000 + [ (145 – 150) * 1 000 000 ]*<br>*New = 457945000+ (-5 * 1 000 000)*<br>*New = 457945000 - 5 000 000*<br>*New = 452945000* |

23. Open HW Config , enter new value at railWidthAxisMaximumPositionLimitInNanometers
24. Click Save at Note Pad.
25. Close V810 Software.
26. Open back V810 Software.
27. Initialize at Panel Handling.
28. Key in“ 150” mm at Panel handling and Click “ Set”
29. Open front access door, use caliper to measure AWA Rail Width & record down the actual measurement and compare with “Parallelism Limit & Width Accuracy Limit “
30. If still not within “Parallelism Limit & Width Accuracy Limit “, repeat Step 1~29 under section “ AWA Parallelism Adjustment” .